

Global Data Analysis

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Introduction

The world database contains rich economic, social, and demographic data that provides valuable insight into global patterns and relationships across countries. To extend this dataset and generate meaningful insights, we incorporated tourism-related economic, employment, and travel pattern data. With this database, the United Nations (UN) can answer key global questions about tourism and make data-driven decisions, supporting its mission to promote responsible, sustainable, and universally accessible tourism (UN Tourism, 2025).

Tourism impacts countries globally. It generates both positive and negative externalities that affect countries economically and socially. Economically, key metrics such as GNP (Gross National Product) and GDP (Gross Domestic Product), which measure a country's overall output of goods and services, allow us to assess tourism's contribution to national economies. However, economic indicators alone are not enough to fully understand tourism's impact. Other factors, such as employment structure and travel patterns, also play a critical role. For example, while tourism may create job opportunities, many of the jobs created may be low-wage food service positions, limiting national income growth. These dynamics highlight how economic growth can come with trade-offs, including social impacts, such as income inequality and economic dependency. The insights derived from this database can help the United Nations evaluate a country's economic stability, employment conditions, and overall quality of life related to tourism, to make more informed, data-driven decisions to support their mission.

The world database includes 239 countries across seven continents and incorporates six tables that support the analysis of tourism patterns and relationships. The central table connecting all others is the Country table, which serves as the foundation of the database. It includes attributes such as name, continent, region, population, life expectancy, GNP, and capital,

providing the geographic, economic, and demographic context needed for analysis. Using country names, continents, regions, and capitals, we can identify where tourism-related impacts are occurring and compare patterns across regions. Life expectancy helps evaluate whether tourism influences overall quality of life, while GNP allows us to assess a country's current economic performance. This table serves as the foundation for all analysis, providing the baseline context needed to interpret tourism data.

The two other base tables, City and CountryLanguage, provide additional detail within each country. The City table includes attributes such as name, district, and population, while the CountryLanguage table lists the languages spoken in a country and identifies which are official. This additional data enables a more detailed, zoomed-in analysis at both the city and cultural levels, providing deeper insight into how tourism may vary within countries. Large capital cities also account for most tourism activity, and understanding whether tourism activity is concentrated in certain urban areas is important. This allows the United Nations to understand how tourism affects different regions and populations within a country, rather than treating each country as a single whole.

The TourismEconomy table is also an essential part of this analysis, as it captures tourism's direct economic impact. It includes attributes such as the percentage of GDP contributed by tourism, as well as domestic and outbound expenditures. This information shows how much, or how little, tourism contributes to a country's economy, and whether it is generating growth or causing economic losses. As a result, this data enables the United Nations to identify whether countries benefit from tourism or experience economic leakage.

The TourismEmployment table is a critical resource that enables a more comprehensive analysis. It includes attributes such as tourism jobs per 1,000 people and food service jobs per

1,000 people. This data provides a more complete view of tourism's impact on society beyond GDP alone. With this perspective, the United Nations can better evaluate how tourism affects employment and overall social conditions. This is important because job creation alone does not guarantee economic well-being, especially if jobs are low-wage or unstable.

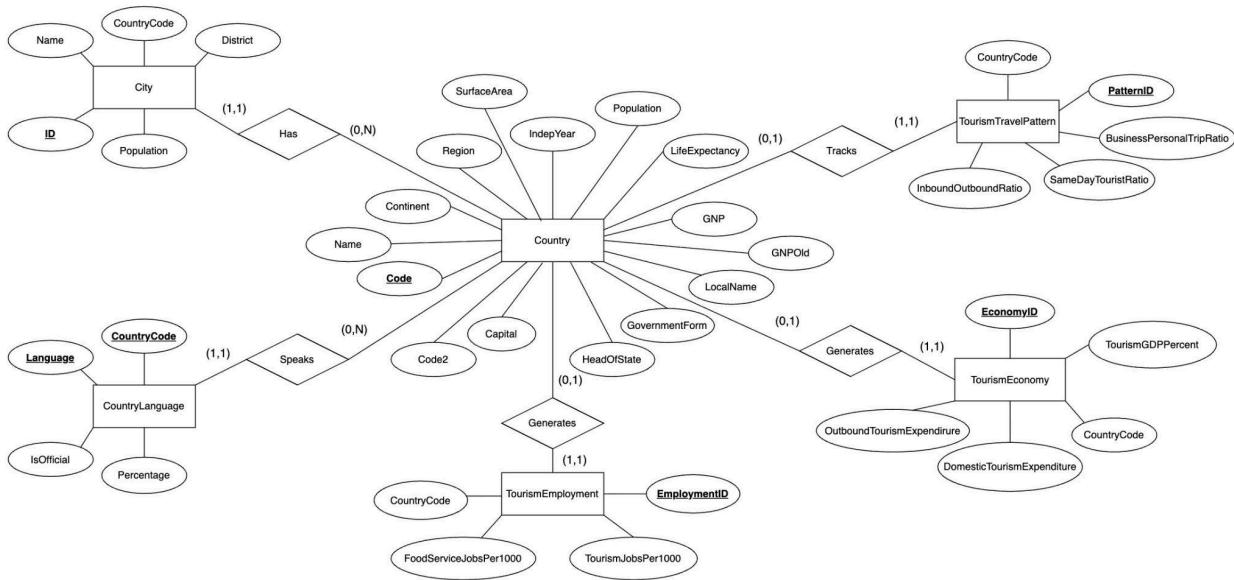
Lastly, the TourismTravelPatterns table includes attributes such as the ratio of business to personal trips and the ratio of inbound to outbound tourists. This data enables us to analyze tourism from a travel behavior perspective and determine whether a country is primarily a tourist destination or if its residents primarily travel to other countries. Understanding these patterns is important because countries that depend heavily on inbound tourism may concentrate their economies in this sector, leading to uneven resource distribution and economic dependency. Analyzing this data enables the United Nations to make more informed decisions when identifying countries that may be overly reliant on tourism, and to assess the long-term sustainability of their economies.

Overall, integrating tourism data into the world database transforms it into a tool for deeper global analysis, supporting one of its essential missions. By combining economic, demographic, cultural, and behavioral data, this database provides a more comprehensive understanding of how tourism impacts countries in different ways. This enables the United Nations to make more informed, data-driven decisions to promote sustainable development and improve the quality of life worldwide.

Database Development Process

Figure 1

Entity-Relationship Diagram



We began the conceptual design phase by examining the initial data in great detail, focusing on entity types, attributes, keys, relationships, and structural constraints. Using the database base tables, we created an entity-relationship diagram (ER Diagram) with 3 core entities: Country, CountryLanguage, and City. Country served as the central entity, with all tables connected to it.

We then identified and classified all attributes, determining whether each was simple, composite, or multivalued; all were determined to be simple (see Figure 1). Once that was completed, we identified all key attributes and underlined them, such as Code in the Country table and ID in the city table. The CountryLanguage entity required a composite key because we determined it was an associative entity, using a combination of its language attribute and the Country entity's primary key to uniquely identify each record.

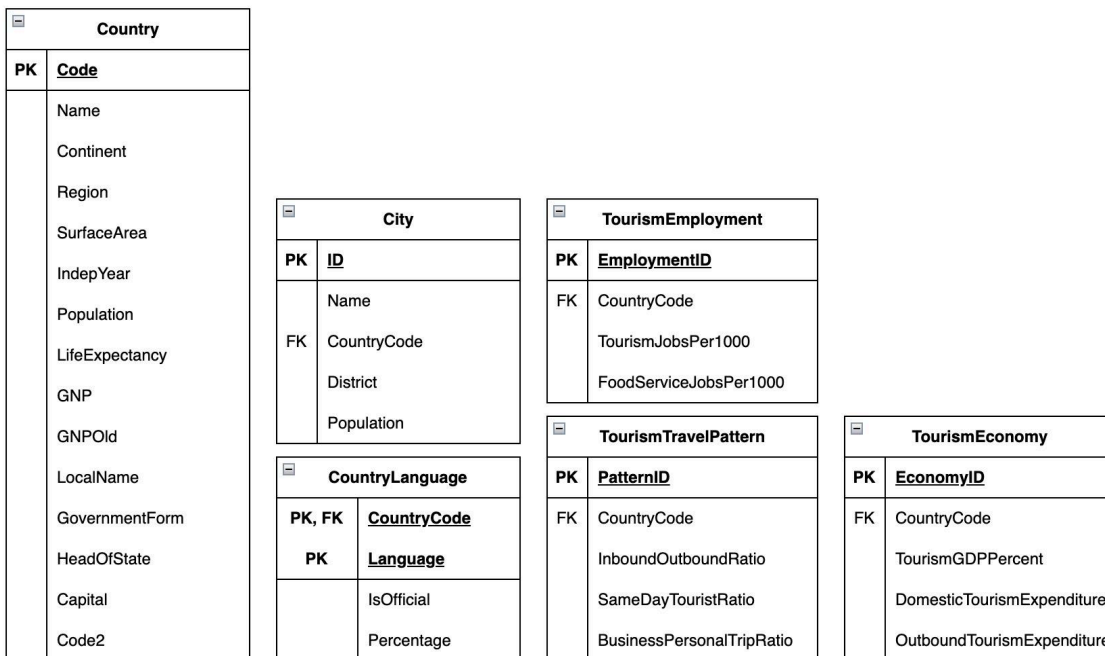
Next, we defined relationships and structural constraints. For example, the HAS relationship between Country and City indicates that a country can have zero to many cities, while each city must belong to exactly one country. After analyzing the data using OUTER JOIN

queries, we found that some countries, such as Antarctica, had no cities recorded, supporting the optional participation of the Country entity in this relationship.

Lastly, we added our three additional tourism tables, TourismEconomy, TourismEmployment, and TourismTravelPatterns, where Country remained the central entity. We applied the same design process to maintain consistency and integrity across the database. These additional entities share similar cardinalities. A country may have zero or one economic, employment, or travel pattern record, because we focused on 1 year of data, and some countries do not have data reported. However, each economy, employment, and travel pattern record must belong to one country. As a structural constraint, the relationship between Country and TourismEmployment would be read as follows: a country may generate no tourism employment and at most one employment record, and tourism employment must be generated by only one country.

Figure 2

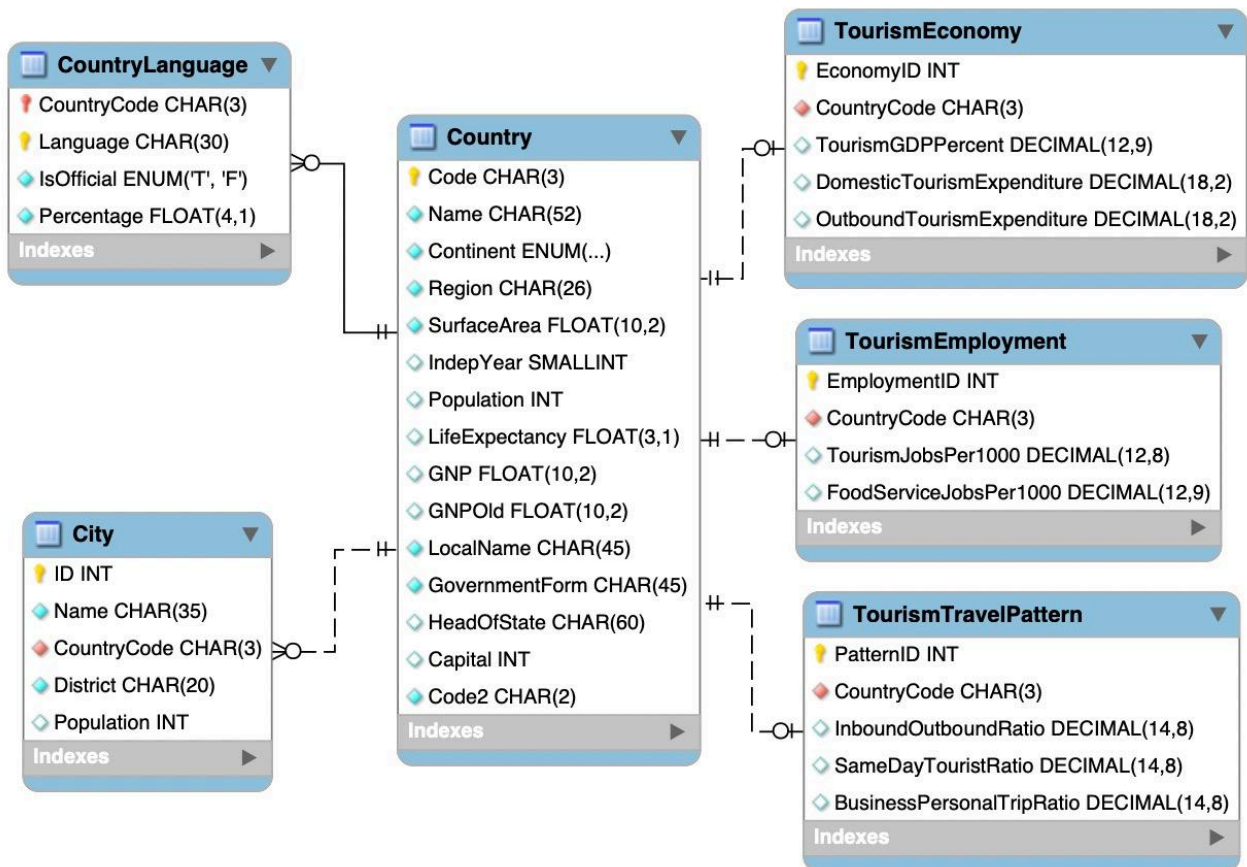
Database Schema



Based on our ER diagram, we designed the schema for the tables and their attributes. This conceptualized how the information would be stored as rows and columns. The additional tourism tables were implemented with relational key constraints, with CountryCode as the foreign key to maintain referential integrity. Each table also includes a surrogate primary key to uniquely identify each record.

Figure 3

Physical Model



Our physical model outlines our database structure. It visualizes the ER diagram's relationships and highlights the central role of the Country table in our database. Additionally, the model uses Crow's Foot notation to represent cardinalities between entities. The model also

shows the data types of each attribute, which are essential for querying and understanding the data. Primary keys, foreign keys, and other constraints, such as null constraints, are also displayed to ensure data integrity and consistency.

Database Investigation

Tourism is a multidimensional activity that affects countries in economic and social ways (Sami, 2024). We create 20 queries to explore these impacts and show various perspectives on key economic and social measures. These queries aim to provide a comprehensive view of how tourism impacts countries within the world database, along with explanations of why this information may be useful for decision-making.

Joins

Query 1

Highest Percentage of GDP Contributed By Tourism.

```
SELECT Name AS CountryName, Continent, TourismGDPPercent  
FROM Country  
JOIN TourismEconomy  
ON code = countrycode  
ORDER BY TourismGDPPercent DESC;
```

This query joins the Country and TourismEconomy tables on the country code. This combines the country name and continent on which it is located with data about how much tourism contributes to each country's GDP. We then sort the results in descending order based on the percentage of GDP from tourism, with countries that rely heavily on tourism appearing at the top. This query provides insight into which countries' economies depend most on the tourism sector. This is important for the UN to further investigate those countries and determine whether

they are seeing positive social benefits as well, since tourism is a multidimensional activity that affects countries in many ways.

Query 2

Tourism GDP Percent and Life Expectancy.

```
SELECT Name AS CountryName, LifeExpectancy, TourismGDPPercent  
  
FROM Country  
  
JOIN TourismEconomy  
  
ON Code = CountryCode  
  
ORDER BY TourismGDPPercent DESC;
```

This query compares each country's life expectancy with the percentage of GDP contributed by tourism. It joins the Country table with the TourismEconomy table on the country code, then orders the results by tourism GDP percentage from highest to lowest. This query is useful for investigating whether countries with greater reliance on tourism tend to have higher life expectancy. Since tourism can increase national income, employment, and infrastructure development, it may contribute to a higher standard of living, which could, in turn, support longer life expectancy. This view is necessary for assessing whether tourism is benefiting or negatively affecting a country.

Query 3

Tourism Jobs Per 1000 And Population.

```
SELECT Name AS CountryName, Population, TourismJobsPer1000  
  
FROM Country  
  
JOIN TourismEmployment  
  
ON code = countrycode
```

ORDER BY Population DESC;

This query joins the Country table with the TourismEmployment table using the country code to link the data. The results are sorted in descending order by population. This allows us to examine the number of tourism jobs in each country against the total population. This query shows which countries depend on tourism as a major source of employment. It is important for the UN to understand a country's economic dependence on tourism to ensure that fluctuations in tourism levels do not negatively affect the nation's people.

Query 4

Percentage of GDP Contributed by Tourism and Tourism Jobs Per 1000.

```
SELECT Name AS CountryName, TourismGDPPercent, TourismJobsPer1000
FROM Country
JOIN TourismEconomy AS TEcon
ON code = countrycode
JOIN TourismEmployment AS TTemp
ON TEcon.countrycode = TTemp.countrycode
ORDER BY TourismGDPPercent DESC;
```

This query joins the Country, TourismEconomy, and TourismEmployment tables on the country code. It then orders the results by the percentage of GDP contributed by tourism, from highest to lowest. This shows whether countries where tourism largely contributes to GDP also have more employment opportunities. This is important to understand to determine whether tourism is creating employment benefits. Job creation can signal positive economic and social impacts, such as individuals earning income from more available jobs and the income they earn improving their quality of life.

Query 5**Inbound/Outbound Tourism and Tourism GDP Percent.**

```
SELECT Name AS CountryName, InboundOutboundRatio, TourismGDPPercent
FROM Country
JOIN TourismTravelPattern AS TTP
ON code = countrycode
JOIN TourismEconomy AS TEcon
ON TTP.countrycode = TEcon.countrycode
ORDER BY InboundOutboundRatio DESC;
```

This query joins the Country, TourismTravelPatterns, and TourismEconomy tables on the country code and sorts the results by InboundOutboundRatio from highest to lowest. It displays the ratio of inbound to outbound tourists, along with the percentage of GDP contributed by tourism. This allows us to evaluate whether countries with higher inbound tourism also experience greater economic benefits. This is important because a country with many inbound tourists but little to no increase in GDP could mean that individuals are not patronizing local businesses or that there is economic leakage. Understanding this is necessary to determine whether increased tourism leads to meaningful economic growth.

Query 6**Capital Cities with a Strong Tourism Economy.**

```
SELECT country.name AS CountryName, city.name AS Capital, city.population AS
CapitalPopulation, TourismGDPPercent
FROM country
JOIN city ON capital = ID
```

```
JOIN tourismeconomy AS TEcon ON code = TEcon.countrycode
ORDER BY TourismGDPPercent DESC;
```

This query joins the Country, City, and TourismEconomy tables on country code, sorting results by TourismGDPPercent in descending order. The output displays the country name, capital city, capital population, and the percentage of GDP contributed by tourism in a single table, showing which countries have the strongest tourism economies and how they relate to major cities. Major cities are major urban centers and typically major tourism hubs where most tourist services are located. This is important to understand whether major cities play a significant role in economic growth related to tourism.

Query 7

Official Languages for Countries with High Tourism GDP Percent.

```
SELECT Name AS CountryName, Language, TourismGDPPercent
FROM Country
JOIN TourismEconomy AS TEcon
ON Code = TEcon.CountryCode
JOIN CountryLanguage AS CL
ON TEcon.CountryCode = CL.CountryCode
WHERE IsOfficial = 'T'
ORDER BY TourismGDPPercent DESC;
```

This query joins three tables, Country, TourismEconomy, and CountryLanguage, on country code. Displayed in a single table are the country name, languages spoken in the country, and the percentage of GDP from tourism. Using the WHERE clause to filter for official languages helps us focus only on official languages recognized at the national level. The results

are then sorted in descending order by tourism GDP percentage, so countries that rely most on tourism appear at the top of the list. Overall, this query is important to understand how language might play a role in tourism success.

Query 8

Countries That Have High Business Travel.

```
SELECT Name AS CountryName, Continent, Region, BusinessPersonalTripRatio  
FROM Country  
JOIN TourismTravelPattern  
ON Code = CountryCode  
ORDER BY BusinessPersonalTripRatio DESC;
```

This query joins the Country table with the TourismTravelPattern table on the country code. This result combines general country information, including name, continent, and region, with travel type data. The results are then sorted in descending order, so the countries at the top show a high level of business travel. BusinessPersonalTripRatio is important because it shows how much of a country's travel activity is driven by business trips versus personal or leisure trips. A higher ratio means more business travel. It is also important to note that if we reversed the order to ascend, we would see countries where personal or leisure travel dominates. Overall, this query helps us understand travel patterns and identify which countries are more business-oriented rather than leisure-driven, which is important for determining whether a country can sustain that level of tourist activity, as personal travel typically indicates a need for more lodging and other tourist services.

Query 9

Tourism GDP Percent and Food Service Jobs.

```
SELECT Name AS CountryName, TourismGDPPercent, FoodServiceJobsPer1000
FROM Country
JOIN TourismEconomy AS TEcon
ON Code = TEcon.CountryCode
JOIN TourismEmployment AS TTemp
ON TEcon.CountryCode = TTemp.CountryCode
WHERE TourismGDPPercent IS NOT NULL AND FoodServiceJobsPer1000 IS NOT
NULL
ORDER BY TourismGDPPercent DESC;
```

This query joins three tables, Country, TourismEconomy, and TourismEmployment, on country code. This allows us to compare tourism's contribution to GDP alongside the number of food service jobs per 1000 people. We also filtered out any missing data to ensure our results are consistent and sorted by tourism GDP percentage in descending order. This lets us analyze whether countries that rely heavily on tourism economically also tend to have more jobs in food service. This helps us explore the connection between tourism's economic impact and its effect on employment. If tourism contributes significantly to a country's GDP and creates many food service jobs, this may suggest an adverse social impact since food service jobs are typically low-wage positions. This is important to understand because this pattern may limit income growth and affect quality of life, contributing to social inequality.

Query 10

Countries Losing Tourism Revenue to Outbound Travel.

```
SELECT Name AS CountryName, Continent, OutboundTourismExpenditure
FROM Country
```

```

JOIN TourismEconomy
ON Code = CountryCode

ORDER BY OutboundTourismExpenditure DESC;

```

This query joins the Country and TourismEconomy table on the country code, then sorts the results by OutboundTourismExpenditure from highest to lowest. The output displays the country name, continent, and outbound tourism expenditure, highlighting which countries spend the most on tourism abroad. The purpose of this query is to display which countries are losing tourism revenue abroad, or experiencing economic leakage. This is a different perspective to determine whether tourism is causing economic loss rather than benefiting the economy. The results also show the continent the country is located within, allowing us to see whether there is a relationship between continents experiencing this issue. This is important in deciding whether intervention is needed to mitigate negative tourism impacts and promote more sustainable tourism development.

Subqueries

Query 1

Tourism GDP Exceeds The Global Average.

```

SELECT Name AS CountryName, TourismGDPPercent
FROM Country
JOIN TourismEconomy ON Code = CountryCode
WHERE TourismGDPPercent > (
    SELECT AVG(TourismGDPPercent)
    FROM TourismEconomy
    WHERE TourismGDPPercent IS NOT NULL);

```

The goal is to identify countries where tourism contributes a significant share of their GDP compared to the global average. The main query will select the country name and its tourism GDP percentage for display. The subquery will perform most of the calculations by computing the average tourism GDP percentage across all countries and returning only countries where their GDP exceeds that average. The results would help us understand that certain countries depend on tourism to support their economies. With such high percentages, we could assume these countries invest more in transportation systems and public services that would benefit both residents and tourists, rather than focusing on in-house production. Furthermore, we could investigate whether tourism investments are associated with improvements in quality of life, like economic stability and life expectancy.

Query 2

High Life Expectancy and Strong Tourism GDP.

```
SELECT Country.Code, Country.Name, Country.LifeExpectancy,
TourismEconomy.TourismGDPPercent
FROM Country
JOIN TourismEconomy
ON Country.Code = TourismEconomy.CountryCode
WHERE Country.LifeExpectancy > (
    SELECT AVG(LifeExpectancy)
    FROM Country
    WHERE LifeExpectancy IS NOT NULL)
AND TourismEconomy.TourismGDPPercent > (
    SELECT AVG(TourismGDPPercent)
```


FROM TourismEconomy

WHERE TourismGDPPercent IS NOT NULL);

This subquery is designed to find countries that are both above average in life expectancy and in tourism GDP percentage. There are several conclusions that could be made from the results. A noticeable pattern is that the majority of the countries are popular global destinations (e.g., Spain, Greece, Italy), and they suggest that tourism can be a stable and beneficial economic pillar when supported by the government and its people. If we take a look at higher life expectancy, it typically reflects better healthcare systems, higher incomes, and important institutions. Additionally, countries that exceed the tourism GDP average often attract many tourists, which allows investment in infrastructure like public facilities, transportation systems, etc. Both of these factors reinforce each other because tourism generates income that improves living conditions, while countries with healthier populations and stable environments are more attractive to tourists, creating a positive cycle supporting economic growth and national well-being.

Query 3

High Tourism Employment and Low Overall GNP.

```
SELECT Country.Name, (Country.GNP / Country.Population) AS GNP_per_capita,
TourismEmployment.TourismJobsPer1000
FROM Country
JOIN TourismEmployment
ON Country.Code = TourismEmployment.CountryCode
WHERE TourismEmployment.TourismJobsPer1000 > (
    SELECT AVG(TourismJobsPer1000)
```

```

FROM TourismEmployment)

AND (Country.GNP / Country.Population) < (

    SELECT AVG(GNP / Population)

    FROM Country

    WHERE GNP IS NOT NULL

    AND Population IS NOT NULL);

```

The purpose of this subquery is to compare each country against two of its averages:

1. TourismJobsPer1000 > average tourism jobs per 1000 people
2. GNP per capita < average GNP per capita

We want to identify countries that are more dependent on tourism jobs than average while still having a lower than average economic output per person. This helps reveal where certain countries can reduce unemployment with tourism being a major employer, but it does not necessarily reduce economic inequality. If we take a look at the tourism industry as a whole, it consists mainly of low-wage service-related jobs such as hotel staff, tour guides, drivers, etc., where, although many people are employed in these jobs, overall national wealth per person will remain relatively low. Another reason for inequality to remain high is called economic leakage. Economic leakage means that tourism revenue leaves the country rather than staying within local communities. Many factors cause this, but one big one is due to foreign-owned businesses. These businesses could be hotels, resorts, or even imported goods that tourists will spend their money on, and the majority of the profits will head overseas instead of supporting the locals.

Query 4

High Inbound Tourism and Low Domestic Tourism Spending.

```
SELECT Country.Name, TourismTravelPattern.InboundOutboundRatio,
```

```
TourismEconomy.DomesticTourismExpenditure
FROM Country
JOIN TourismTravelPattern
ON Country.Code = TourismTravelPattern.CountryCode
JOIN TourismEconomy
ON Country.Code = TourismEconomy.CountryCode
WHERE TourismTravelPattern.InboundOutboundRatio > (
    SELECT AVG(InboundOutboundRatio)
    FROM TourismTravelPattern
    WHERE InboundOutboundRatio IS NOT NULL)
AND TourismEconomy.DomesticTourismExpenditure < (
    SELECT AVG(DomesticTourismExpenditure)
    FROM TourismEconomy
    WHERE DomesticTourismExpenditure IS NOT NULL);
```

The goal for this subquery is to highlight countries that depend more on foreign visitors than on spending by their residents. To be more specific, these countries likely rely more on foreign tourism demand than on internal markets. When inbound tourism is high, but domestic tourism expenditure is low, we can assume that residents have lower disposable income for tourism, meaning lower purchasing power, and tourism infrastructure is centered on foreigners. Having lower disposable incomes means people have less money left after paying for essentials. Since tourism is considered a “luxury” expense, it is often a situation where households will hold off when budgets are tight. We must note that international tourists often come from countries with higher wages and stronger currencies. So even if prices may seem normal to visitors, they

may feel expensive to residents earning minimum wage. Although the economy is supported by tourism, it's also a major flaw. They are dependent on a single source of demand, and tourism is highly sensitive to external events such as recessions, pandemics, or political instability. Once these events happen, countries that rely on visitors will experience sharp losses with greater instability and much slower economic recovery.

Query 5

Above Average Domestic Tourism Expenditure.

```
SELECT Country.Name, TourismEconomy.DomesticTourismExpenditure
FROM Country
JOIN TourismEconomy
ON Country.Code = TourismEconomy.CountryCode
WHERE TourismEconomy.DomesticTourismExpenditure > (
    SELECT AVG(DomesticTourismExpenditure)
    FROM TourismEconomy
    WHERE DomesticTourismExpenditure IS NOT NULL);
```

The goal is to identify countries with strong internal tourism demand, indicating economic strength and development. By comparing each country's domestic tourism spending against the average, we can isolate countries whose tourism industries are not only supported by visitors, but also by active participation from their own populations. Similar to the previous case we examined, we are now looking at countries that reflect higher disposable income. Countries with above-average domestic spending usually have higher household income and larger middle-upper-class populations. Residents must meet essential needs before traveling. This further suggests stronger purchasing power among citizens, and the country likely has a

well-developed infrastructure where traveling is convenient and accessible. Countries that support domestic and international travelers are more resilient and sustainable in the long term, as tourism is often seasonal and domestic tourism provides a consistent year-round demand.

Query 6

Countries Where Tourism is A Major Source of Employment.

```
SELECT Country.Name, Country.Population, TourismEmployment.TourismJobsPer1000
FROM Country
JOIN TourismEmployment
ON Country.Code = TourismEmployment.CountryCode
WHERE TourismEmployment.TourismJobsPer1000 > (
    SELECT AVG(TourismJobsPer1000)
    FROM TourismEmployment
    WHERE TourismJobsPer1000 IS NOT NULL)
ORDER BY TourismEmployment.TourismJobsPer1000 DESC;
```

This query identifies the countries where the tourism employment per 1,000 residents exceeds the global average. By filtering against the average, the results isolate countries where tourism plays a significant role in job creation. When examining employment rates relative to population sizes across countries, the results reveal that countries of varying sizes can consistently achieve high employment rates. This insight is crucial for assessing countries worldwide, as it reveals which countries offer the most tourism opportunities and which are most at risk if tourism activity declines.

Query 7

Tourism Employment in the Most Populous Countries.

```
SELECT Country.Name, TourismEmployment.TourismJobsPer1000
FROM Country
JOIN TourismEmployment
ON Country.Code = TourismEmployment.CountryCode
WHERE TourismEmployment.TourismJobsPer1000 > (
    SELECT AVG(TourismJobsPer1000)
    FROM TourismEmployment)
AND CountryCode IN (
    SELECT Code
    FROM Country
    WHERE Population > (
        SELECT AVG(Population)
        FROM Country));
```

This query focuses on the larger countries with a high tourism employment rate. The goal was to identify the countries that were globally above average in both population size and tourism employment rate. The larger countries have diversified economies, and this can reveal the performance of tourism alongside other major industries in generating jobs at a national scale. The results suggest that tourism employment does not naturally scale with population size, and that returns are greatest in countries where tourism investment or workforce development programs could have the broadest reach and impact.

Query 8

Countries with Greater Inbound Tourists than Outbound.

```
SELECT Country.Name, TourismTravelPattern.InboundOutboundRatio
```

```

FROM Country
JOIN TourismTravelPattern
ON Country.Code = TourismTravelPattern.CountryCode
WHERE TourismTravelPattern.InboundOutboundRatio > 1
ORDER BY TourismTravelPattern.InboundOutboundRatio DESC;

```

Query 8 filters the TravelPattern table to find countries where the inbound to outbound tourist ratio exceeds 1, meaning more foreign visitors arrive than residents travel abroad. This compares a country's travel habits with the number of tourists it attracts, indicating strong destination appeal. The extreme inbound ratio doesn't inherently mean a country's economy is exceeding, but rather it draws attention to the economies that benefit the most from tourism.

Query 9

Linguistic Diversity and Tourism Employment.

```

SELECT Country.Name, TourismEmployment.TourismJobsPer1000
FROM Country
JOIN TourismEmployment
ON Country.Code = TourismEmployment.CountryCode
WHERE Country.Code IN (
    SELECT CountryCode
    FROM CountryLanguage
    WHERE IsOfficial = 'T'
    GROUP BY CountryCode
    HAVING COUNT(Language) > 1)
ORDER BY TourismEmployment.TourismJobsPer1000 DESC;

```

This query finds all countries that have more than one official language and ranks them by their employment figures. Our goal was to determine the relationship between tourism activity and linguistic diversity. By examining tourism employment figures across the group of countries with more than 1 official language, the role of language can be measured in terms of a country's ability to generate tourism-related jobs. The results did not prove linguistic diversity to be a reliable predictor of employment; however, it does factor into the performance by providing language accessibility. The higher-ranking countries tended to perform moderately well in tourism employment.

Query 10

Tourism Economic Contribution in Countries with Major Capital Cities.

```

SELECT Country.Name, Country.Capital, TourismEconomy.TourismGDPPercent
FROM Country
JOIN TourismEconomy
ON Country.Code = TourismEconomy.CountryCode
WHERE Country.Capital IN (
    SELECT ID
    FROM City
    WHERE City.Population > (
        SELECT AVG(Population)
        FROM City
        WHERE Population IS NOT NULL))
AND TourismEconomy.TourismGDPPercent > (
    SELECT AVG(TourismGDPPercent)

```



```
FROM TourismEconomy
```

```
WHERE TourismGDPPercent IS NOT NULL)
```

```
ORDER BY TourismEconomy.TourismGDPPercent DESC;
```

This query calculates the average population and tourism GDP globally to filter the countries that exceed both conditions. The results reveal countries with infrastructure capable of serving high volumes of tourists. Additionally, the query suggests that the size of the capital city may matter less than other factors that draw tourists. This information can be useful for strengthening tourism economies. By examining the countries that ranked highest, developing nations can evaluate how to invest in urban infrastructure.

Summary

The development of this global tourism database was not linear and required careful planning and adjustment. Our team began by reviewing the provided country- and city-level data, identifying the key insights we wanted to generate, finding relevant datasets, and selecting appropriate attributes before building the database around those goals. Throughout this process, we encountered several challenges, including the fact that real-world data is often incomplete, contains many NULL values, and includes a wide range of attributes that require careful selection. These impediments directly affect analysis accuracy, making data selection critical.

One major challenge was selecting an appropriate time frame. Initially, we decided to include a “Year” attribute, which ranged from 1995 to 2021. In determining which years to use, we were conscious that the COVID-19 pandemic in 2020 significantly disrupted global tourism, making that data unreliable, while older data had become less relevant. To maintain accuracy and consistency, we narrowed the dataset to a single year, 2018. This ensured the data was both reliable and comparable across tables while still reflecting recent, pre-pandemic trends.

During the database design phase, JOIN operations helped us better understand the relationships between tables. In particular, OUTER JOINs showed that some countries did not have corresponding records in certain tables, highlighting gaps in the data. This informed our ER Diagram but also revealed inconsistencies in real-world data collection, reinforcing the importance of handling missing relationships in database systems.

Tourism, by definition, unites countries by allowing people to travel across borders for cultural exploration, business, and personal reasons. Because of this, understanding global tourism patterns is important for the United Nations, as it helps identify trends, evaluate economic impacts, and determine where support or intervention may be needed to address overtourism and its effects. By designing and implementing a database that integrates country, city, language, and tourism economic, employment, and travel pattern data, along with analytical queries, the United Nations can better understand the role tourism plays across countries and use that information to guide collaboration, improve quality of life, and fulfill its mission.

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